Homework

Homework 14 — Quality of an estimate — 55 min.

We repeat $n \geq 1$ independent Bernoulli trials with the same parameter $p \in [0,1]$, which is unknown. Let F_n be the frequency of successes.

1. Prove that $E(F_n) = p$ and $V(F_n) = \frac{p(1-p)}{n}$.
2. Show that $p(1-p) \leq \frac{1}{4}$ for all $p \in [0,1]$.
3. Deduce a size n sufficient for $P(|F_n - p| < 0.02) \geq 0.95$, whatever the value of p.
4. We now know that $p \leq 0.1$. Determine a sharper upper bound for $p(1-p)$, then improve the sufficient size obtained in question 3.
5. For this question only, the n results are all equal to the same Bernoulli variable B with parameter p. Calculate F_n and its variance. Explain why the previous guarantee no longer applies.